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The puckering free-energy surface of proline

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Proline has two preferred puckering states, which are often characterized by the pseudorotation phase angle and amplitude. Although proline's five endocyclic torsion angles can be utilized to calculate the phase angle and amplitude, it is not clear if there is any direct correlation between each torsion angle and the proline-puckering pathway. Here we have designed five proline puckering pathways utilizing each torsion angle χ_j ($j = 1\sim5$) as the reaction coordinate. By examining the free-energy surfaces of the five puckering pathways, we find they can be categorized into two groups. The χ_2 pathway (χ_2 is about the C_β—C_γ bond) is especially meaningful in describing proline puckering: it changes linearly with the puckering amplitude and symmetrically with the phase angle. Our results show that this conclusion applies to both *trans* and *cis* proline conformations. We have also analyzed the correlations of proline puckering and its backbone torsion angles ϕ and ψ. We show proline has preferred puckering states at the specific regions of ϕ, ψ angles. Interestingly, the shapes of ψ-χ_2 free-energy surfaces are similar among the *trans* proline in water, *cis* proline in water and *cis* proline in the gas phase, but they differ substantially from that of the *trans* proline in the gas phase. Our calculations are conducted using molecular simulations; we also verify our results using the proline conformations selected from the Protein Data Bank. In addition, we have compared our results with those calculated by the quantum mechanical methods. [http://dx.doi.org/10.1063/1.4799082]

I. INTRODUCTION

Proline is a special amino acid because its side chain composes a pyrrolidine ring structure that directly influences its backbone conformations. The pyrrolidine ring prefers a puckered state to relax from the strained planar conformation. Usually, proline is found in two preferred puckering states as shown in Figs. 1(A) and 1(B), where the five endocyclic torsion angles χ_j ($j = 1\sim5$) are labeled as well. The puckered conformation is called UP (C_γ exo, $-\chi_1$, $+\chi_2$, $-\chi_3$) if the C_γ atom and the carbonyl group lie on the opposite side of the plane defined by the ring atoms N, C_α, C_β and $C_{\delta,}$ and it is called DOWN (C_γ endo, $+\chi_1$, $-\chi_2$, $+\chi_3$) if they lie on the same side of the plane.[1] The two puckered states both are populated when analyzing proline conformations from the PDB structures.[2–4] However, at the specific regions, proline's puckering state can affect the stability of the protein structures. For example, the collagen structure usually contains Xaa-Yaa-GLY unit repeats. L-Proline and 4(*R*)-hydroxy-L-proline are often found at positions of Xaa and Yaa where they prefer the DOWN and UP puckering states, respectively.[3,5] Thus, the pyrrolidine ring puckering states together with other factors[6] are found useful to interpret the collagen structure stability. In addition, the process of proline puckering is also found meaningful. Juffer and co-workers have studied the proline conformational transition from a planar to a puckered ring structure, which is associated with the conformational switch when the substrate is released from the triosephosphate isomerase.[7]

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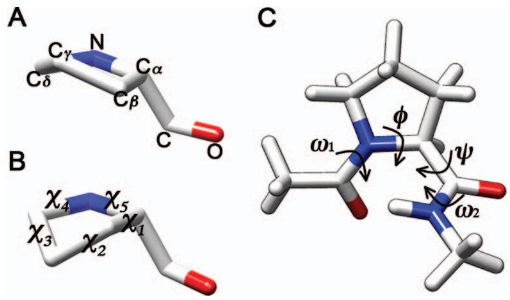

FIG. 1. Schematic pictures of the proline model. The five ring-atom names and the five endocyclic torsion angles χ_j are labeled. (A) A typical UP puckering conformation, also called the C_γ exo conformation, which is manifested by the negative χ_1, positive χ_2, and negative χ_3 values. (B) A typical DOWN puckering conformation (C_γ endo, $+\chi_1$, $-\chi_2$, $+\chi_3$). (C) Proline dipeptide with the torsion angles ϕ, ψ, ω_1 and ω_2 labeled. Hydrogen atoms are shown in (C) but are omitted in (A) and (B). The graphs are visualized using Chimera.[38]

As an important and special feature of proline, the pyrrolidine ring puckering has been studied in detail,[4,8–17] and proline's two preferred puckering states have been characterized very well. Usually, the pyrrolidine ring puckering is symbolized with two parameters—the pseudorotation phase angle and amplitude—that can be calculated either by the five ring-atom positions[18] or by the five endocyclic torsion angles χ_j.[19,20] Quite often, the five χ_j values are calculated for each puckering state and are listed as part of the puckering parameters in the results.[12,15,16] It is easy to see that the sign of several χ_j switches at the UP and DOWN puckering states, respectively. Madison has found that at the two puckering states, the magnitudes of χ_1, χ_2 and χ_3 are large whereas those of χ_4 and χ_5 are small.[9] These properties of χ_j have been utilized when studying proline puckering, and the sign of χ_1 has often been utilized to symbolize the two puckering states.[2–4] All these studies demonstrate that the five torsion angles are important in characterizing proline puckering. However, the current usage of χ_j only limits to the symbolization of two stationary puckering states (characterized with the positive or negative χ_j values), whereas the proline puckering along the full path of χ_j has not been studied systematically. For example, how proline puckering depends on the variation of each χ_j is not clear. In addition, less work has focused on the study of χ_4 and χ_5 in relation to proline puckering, which needs to be elucidated further.

In this study, we aim to address the following questions: (1) How does proline puckering depend on the variation of each χ_j? (2) What are the relations among the five χ_j? (3) What is the difference between the proline puckering characterized with χ_j and that characterized with the pseudorotation phase angle and amplitude? To answer these questions, we have specifically designed five proline puckering pathways using each χ_j as the reaction coordinate. We have calculated the free-energy surfaces of proline puckering along each χ_j using the puckering parameters—pseudorotation phase angle and amplitude. We have also calculated the puckering free-energy surfaces represented by any two χ_j to further elucidate the relations among the five χ_j. In addition, we have compared the free-energy surfaces of proline puckering along the backbone torsion angles ϕ and ψ using the representative χ_j with those using the pseudorotation puckering parameters, and we have identified the ϕ, ψ regions that prefer the specific puckering state. Because the *trans* and *cis* isomerization states both are important when studying proline conformations, we have studied proline puckering at these two states in the gas phase and in water, respectively.

Previously, proline conformations have been studied by molecular simulations using different force fields and by quantum mechanical (QM) methods calculated at the different levels of theory. Karplus and co-workers have utilized the Charmm force field to study proline isomerization[21] and the proline puckering of a cyclic decapeptide.[10] McCammon, Hamelberg, and co-workers have utilized the Amber force field combined with the accelerated molecular dynamics simulations to study proline isomerization and the enzymatic prolyl isomerization process.[22] Doshi and Hamelberg have also modified the Amber force field parameters to study proline isomerization so that the simulation results compared well with the experimental values.[23] Sieradzan, Scheraga and Liwo have utilized a coarse-grained UNRES force field to study the isomerization process.[24] As to the

QM calculations, Kang and co-workers have thoroughly studied the preferable proline conformations using the different QM calculations,[11,12,14,16,25] and in a recent work Kang and Byun have compared the various QM calculations in studying the conformations of alanine dipeptide and proline dipeptide in the gas phase and in water.[26] Aliev and co-workers have studied the proline conformations using several QM calculations,[17] and they have also compared the results of NMR, DFT, and the empirical force field calculations.[15] Sahai *et al*. have studied the four forms proline conformations, differing in *N*- and *C*-terminals, using the QM methods calculated at three levels of theory.[13] Although the preferred proline conformations have been described quite well by these studies, the detailed pyrrolidine puckering pathway along each endocyclic torsion angle and the relations among the five pathways are not clear. The two-dimensional plots of the *trans* proline puckering along the backbone torsion angles (ϕ-χ_1 and ψ-χ_1) have been obtained by analyzing the PDB structures,[2–4] however, the ϕ-χ_j and ψ-χ_j free-energy surfaces of the *cis* proline puckering still need to be elucidated due to the inadequate *cis* proline conformations in the PDB structures. Here we utilize Charmm22 force field[27] (with the CMAP corrections)[28] combined with the Distribution method[29] to study the pyrrolidine ring puckering of both *trans* and *cis* proline conformations. We choose Charmm22 force field because the proline puckering motion had been studied using it and the results were compared well with the experimental values.[10] In this study, we provide a full comparison of the proline structural data at the different states under different conditions (*trans* and *cis* conformations, in the gas phase and in water) calculated by the simulation method with those calculated by the different QM studies. Because the pyrrolidine ring puckering depends on both ϕ and ψ, these comparisons provide a clear view of the relations among ϕ, ψ and χ_j at the different states.

In Sec. II, we introduce the methods that are often utilized to characterize proline puckering, and we explain our methods in Sec. III. Then we present the results in Sec. IV and provide the discussion in Sec. V. Finally, we give a short conclusion in Sec. VI.

II. THEORY

Ring puckering needs to be characterized, which is not a simple task. It is desired to use simplified parameters that can describe the movements of the ring atoms. Phase angle and amplitude are such parameters that can describe the five-membered ring puckering very well,[18–20,30] and they are widely utilized nowadays. (Note additional parameters are needed to describe more than five-membered ring puckering).[18] There are several versions of calculating the ring puckering parameters over a long period of time, demonstrating the continuing efforts of developing and using the reduced parameters to characterize a puckered ring. In 1947, Kilpatrick, Pitzer and Spitzer proposed to use the wave function to describe the cyclopentane ring puckering:[30]

$$z_j = (2/5)^{1/2}\, q_m \cos\left[p_m + 4\pi(j-1)/5\right] \tag{1}$$

Here z_j describes the *j*th ring-atom's displacement perpendicular to the plane of an unpuckered ring, where j takes the value from 1 to 5. p_m is the phase angle and q_m is the puckering amplitude. Thus, the two parameters p_m and q_m can be utilized to describe the cyclopentane ring puckering. Note in the original paper, φ was used to describe the phase angle ($\varphi = p_m/2$), and j instead of ($j-1$) was used in the formula.[30] Although Eq. (1) is accepted to describe the ring puckering, it is not easy to define the generalized unpuckered-ring plane.

Then in 1972, Altona and Sundaralingam proposed to use the endocyclic torsion angles to calculate the phase angle and amplitude that can be used to describe the five-membered ring puckering.[19] Similar to Eq. (1), any of the five torsion angles can be calculated from the phase angle and amplitude; the phase angle and amplitude, in turn, can be calculated by the five torsion angles:

$$\chi'_j = Q_m \cos\left[P_m + 4\pi(j-1)/5\right] \tag{2a}$$

$$\tan P_m = \frac{(\chi'_3 + \chi'_5) - (\chi'_2 + \chi'_4)}{2\chi'_1(\sin \pi/5 + \sin 2\pi/5)} \tag{2b}$$

When Eq. (2) is applied to calculate proline puckering, χ_1' is often chosen to be the torsion angle about the C_β - C_γ bond, so that: $\chi_1' = \chi_2$, $\chi_2' = \chi_3$, $\chi_3' = \chi_4$, $\chi_4' = \chi_5$, $\chi_5' = \chi_1$. Here, we have utilized the capital letters P_m and Q_m to emphasize that the calculations are conducted using the torsion angles instead of the ring-atom positions. Note 180° is added to P_m if $\chi_1' < 0°$. It is argued that Eq. (2a) is only an approximation[18] although it is convenient to use.

In 1975, Cremer and Pople proposed a procedure for the calculation of the ring puckering parameters using the ring-atom positions only.[18] Their derivations do not involve approximations, and the results are generally applied to characterize the N-membered ring puckering, where $N > 3$. Here we present their equations in calculating the five-membered ring puckering only:

$$q_m \cos p_m = (2/5)^{1/2} \sum_{j=1}^{5} z_j \cos\,[4\pi(j-1)/5] \tag{3a}$$

$$q_m \sin p_m = -\,(2/5)^{1/2} \sum_{j=1}^{5} z_j \sin\,[4\pi(j-1)/5] \tag{3b}$$

Here, p_m and q_m can be calculated by Eqs. (3a) and (3b). Following their derivations, the displacement z_j is described exactly by Eq. (1) that was defined by Kilpatrick and co-workers. Note in describing more than five-membered ring puckering, additional puckering parameters are needed, which are included in the Cremer and Pople equations.[18]

In 1981, Rao, Westhof, and Sundaralingam proposed to use the approximate Fourier analysis to describe the five-membered ring puckering using the endocyclic torsion angles χ_j':[20]

$$A = Q_m \cos P_m = (2/5) \sum_{j=1}^{5} \chi_j' \cos\,[4\pi(j-1)/5] \tag{4a}$$

$$B = Q_m \sin P_m = -\,(2/5) \sum_{j=1}^{5} \chi_j' \sin\,[4\pi(j-1)/5] \tag{4b}$$

Like Eq. (3), the two puckering parameters Q_m and P_m can be calculated using Eqs. (4a) and (4b). For example: $Q_m = \sqrt{A^2 + B^2}$ and $P_m = \tan^{-1}(B/A)$. And like Eq. (2), 180° is added to P_m if $\chi_1' < 0°$.

In general, the ring puckering parameters can be calculated either by the ring-atom positions using Eq. (3) or by the endocyclic torsion angles using Eq. (2) or Eq. (4). Although Eq. (2a) is only an approximation,[18] the deviation is small and the results are analogous to those calculated by Eq. (3). Harvey and Prabhakaran have shown when describing the ribose ring puckering, p_m and P_m are linearly correlated at each puckering state although their values are not exactly same.[31] In addition, q_m and Q_m are also related, and their relation can be approximated by a simple equation:[32]

$$Q_m(\mathrm{deg}) = 102.5 q_m(\text{Å}) \tag{5}$$

Here the units of q_m and Q_m are different. On the other hand, the differences between the puckering parameters calculated by Eqs. (2) and (4), both utilizing the endocyclic torsion angles, are small and can be neglected.[31]

The endocyclic torsion angles are useful in the calculation of the ring puckering parameters. They are utilized not only for describing the five-membered ring puckering, but also for the six-membered ring puckering.[33] A recent development has focused on utilizing a combination of conformations represented by the torsion angles to describe the six-membered ring puckering.[34] In our study, we find that if directly using two torsion angles to describe proline puckering, it is analogous to using the triangular decomposition method proposed by Hill and Reilly.[35] In their method, N – 3 parameters are utilized to describe the N-membered ring puckering. For example, the pyrrolidine ring can be divided into three subtriangular planes, and the puckering is described by the two dihedral angles τ_j along the two adjacent subplanes (Fig. 2, see Method section for the detailed descriptions). Here, we find the dihedral angles τ_j have the close relationship with the endocyclic torsion angles χ_j, so that the free-energy surfaces described by two τ_j are analogous to those described by two χ_j.

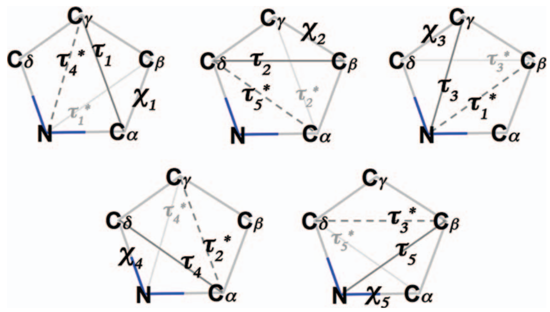

FIG. 2. The five schemes of dividing the pyrrolidine ring structure using the two dihedral angles τ_j (along the *solid* ***dark**-gray* line) and τ_k* (along the *dashed* ***dark**-gray* line), which are analogous to the puckering parameters utilized in the triangular decomposition method. Proline's endocyclic torsion angle χ_j and the dihedral angle τ_j* (along the *solid* ***light**-gray* line) are also labeled in each ring structure because of the close relations among the three values τ_j, τ_j* and χ_j (described in the Method section). Here we define τ_1: N-C_α-C_γ-C_β, τ_2: C_α-C_β-C_δ-C_γ, τ_3: C_β-C_γ-N-C_δ, τ_4: C_γ-C_δ-C_a-N, τ_5: C_δ-N-C_β-C_a, τ_1*: C_γ-C_β-N-C_α, τ_2*: C_δ-C_γ-C_α-C_β, τ_3*: N-C_δ-C_β-C_γ, τ_4*: C_α-N-C_γ-C_δ, τ_5*: C_β-C_α-C_δ-N.

III. METHOD

We choose proline dipeptide (Fig. 1(C)) as the model system and use molecular simulations to study proline puckering. We have used the Charmm22 force field parameters[27] and included the CMAP corrections[28] in the calculations. In the aqueous-phase simulations, proline dipeptide is immersed in a cubic box filled with 494 TIP3P water molecules. The peptide-water and water-water interactions both are truncated at 9.5 Å, with the box length fixed at 25 Å and the temperature fixed at 298K. In order to check if the truncation we applied is suitable, we have conduced separate simulations for the *trans* proline puckering in water using the truncation of 12 Å. We find that the free-energy surface of the χ_2 puckering pathway does not change and the structural data (ϕ, ψ, χ_j, ω_1, ω_2, as labeled in Fig. 1(C)) are also consistent with those calculations when the truncation is set to 9.5 Å. Therefore, we have chosen the truncation of 9.5 Å for the aqueous-phase calculations. The free-energy surfaces are calculated using the Distribution Method.[29] The detailed implementation of the distribution method for the proline dipeptide system is similar to that of the alanine dipeptide system as described in Ref. 36. When calculating the free-energy surfaces, the reaction coordinates are chosen from a list of five endocyclic torsion angles χ_j and the proline backbone torsion angles ϕ and ψ. In this study, when calculating the χ_j pathways, χ_j, P_m and p_m all are binned at 6°, Q_m is binned at 2°, and q_m is binned at 0.02 Å. When calculating ϕ-χ_j and ψ-χ_j free-energy surfaces, ϕ, ψ and χ_j all are binned at 6°. We have also checked the convergence of the *trans* proline χ_2 pathway calculations. The average relative errors of the energy calculations (the absolute error divided by the average total energy of the system) conducted in the gas phase and in water both are less than 0.1%. The absolute average errors of the free-energy calculations in the gas phase and in water are 0.01 kcal/mol and 0.05 kcal/mol, respectively. These data show that the χ_2 pathway calculations are converged. Because we have used the same type of calculations for all the five χ_j pathway simulations (they only differ in the reaction coordinate χ_j), the convergence of the calculation along the other χ_j pathways should not differ too much from that of the χ_2-pathway calculations.

To verify the simulation results, we have also calculated the free-energy surfaces using the proline conformations selected from the Protein Data Bank (http://www.rcsb.org). The PDB files are filtered using the following rules: structures solved by only X-ray method are selected, with the resolution < 1.7 Å, R-factor < 0.2, B-factor < 40, sequence identity < 50%; residues missing atoms or having alternative atom positions are not selected. Here we only select proline conformations having $\omega_1 < -90°$ or $\omega_1 > 90°$, and $\omega_2 < -90°$ or $\omega_2 > 90°$, where ω_1 and ω_2 are as labeled in Fig. 1(C). This means we only select proline residues having both ω_1 and ω_2 in the *trans* conformations. Following these rules, we have selected 1572 PDB files and 13797 proline residues for the study. We do not analyze PDB structures having *cis* proline conformations because their numbers are small and are not adequate for the accurate description of the free energy surfaces.

Generally, the monocyclic ring puckering has $N-3$ degrees of freedom.[18] Hence the pyrrolidine ring puckering can be described by two parameters. In the triangular decomposition method,[35] Hill and Reilly have chosen the dihedral angles that divide the ring plane to describe the ring puckering. In this study, we show that this method is analogous to using the two endocyclic torsion angles χ_j when characterizing the proline puckering. Figure 2 shows the five schematic pictures of the pyrrolidine ring, each of which is divided into three subplanes by the *dashed* and the *solid **dark**-gray* lines. The two adjacent subplanes can define one dihedral angle, so there are two dihedral angles in each pyrrolidine ring structure. In each ring, we denote the dihedral angle along the *solid **dark**-gray* line as τ_j, and denote the dihedral angle along the *dashed **dark**-gray* line as $\tau_k{}^*$. Note $\tau_k{}^*$ can also be found in the other subplot, where it is denoted as $\tau_j{}^*$ along the *solid **light**-gray* line. For example, in the upper-left subplot of Fig. 2, the two dihedral angles, dividing the pyrrolidine ring into three subplanes, are τ_1 and $\tau_4{}^*$, respectively. And $\tau_4{}^*$ can also be found in the lower-left subplot. We draw $\tau_j{}^*$ (along the *solid **light**-gray* line) in each subplot because we find from our calculations that $\tau_j \approx \tau_j{}^* \approx \chi_j + 180^\circ$, where τ_j, $\tau_j{}^* \in [0^\circ, 360^\circ]$, $\chi_j \in [-180^\circ, 180^\circ]$, and $j = 1 \sim 5$. We note the correlations among χ_j, τ_j and $\tau_j{}^*$ are good when the puckering amplitude is small, but the differences among the three values increase as the puckering amplitude becomes large. Generally, their differences are within $\pm 6^\circ$ when the puckering amplitude is less than 60°. This means when describing the puckering free-energy surface of proline, using any two dihedral angles τ_j and $\tau_k{}^*$ is similar to using any two corresponding torsion angles χ_j and χ_k. Note Hill and Reilly originally utilized θ in the triangular decomposition method.[35] Here τ_j can be related to θ_j with the equation: $\theta_j = 180^\circ - \tau_j$, where $\theta_j \in [-180^\circ, 180^\circ]$.

In the following calculations, we use the different puckering parameters to describe the free-energy surfaces of proline puckering. The phase angle and amplitude p_m and q_m are calculated using Eq. (3) with C_γ defined as the first atom ($j = 1$); P_m and Q_m are calculated using Eq. (4) with χ_2 defined as the first torsion angle. We have also utilized the combination of two χ_j or two τ_j to describe proline puckering free-energy surfaces, so that we can compare the results calculated by any two τ_j with those calculated by the corresponding two χ_j.

Conventionally, UP and DOWN are often utilized to denote proline's two preferred puckering states. In this work, we prefer to use the sign of χ_2 to symbolize these two states because it is convenient to find the sign of χ_2 in each free-energy plot. In the following sections, **positive-χ_2** and **negative-χ_2** are utilized to denote the UP and DOWN puckering states, respectively.

In our calculations, we have studied both the *trans* and *cis* conformations of the proline dipeptide via the simulation method. Here the ***trans* proline conformation** refers to both ω_1 and ω_2 are in the *trans* conformations, and the ***cis* proline conformation** refers to ω_1 in the *cis* conformation and ω_2 in the *trans* conformation.

IV. RESULTS

A. The relation between the torsion angle χ_j and the phase angle p_m or P_m

In order to illustrate the correlation between each χ_j and proline puckering, we have designed five pathways, in each of which χ_j value changes continuously from -60° to 60°. We call the proline puckering described in each pathway the "χ_j-puckering pathway" or simply the "χ_j pathway." In Fig. 3, we plot the free-energy surfaces of χ_j-p_m and χ_j-P_m of *trans* proline (both ω_1 and ω_2 are in *trans* conformations) in the gas phase. The unit of the free energy plotted in Fig. 3 is kcal/mol that applies to all the following figures as well. The two bell-shaped regions delineated by the wave curve describe the pyrrolidine ring puckering deviating from the generalized unpuckered ring. The phase angle (p_m or P_m) expands a wide area as χ_j approaches 0°, but its value converges as χ_j approaches -60° or 60°. In the five χ_j pathways, proline's two preferred puckering states do not change and are characterized by the two dark-blue regions in each subplot. However, the gas-phase results show proline prefers one puckering state to the other. Take χ_2 pathway for an example, proline prefers the negative-χ_2 puckering state to the positive-χ_2 puckering state, manifested by the darker blue region (lower free-energy values) and the lighter blue region (higher free-energy values). This conclusion holds for all χ_j pathways. Comparing the shapes of the two puckering states along each χ_j pathway,

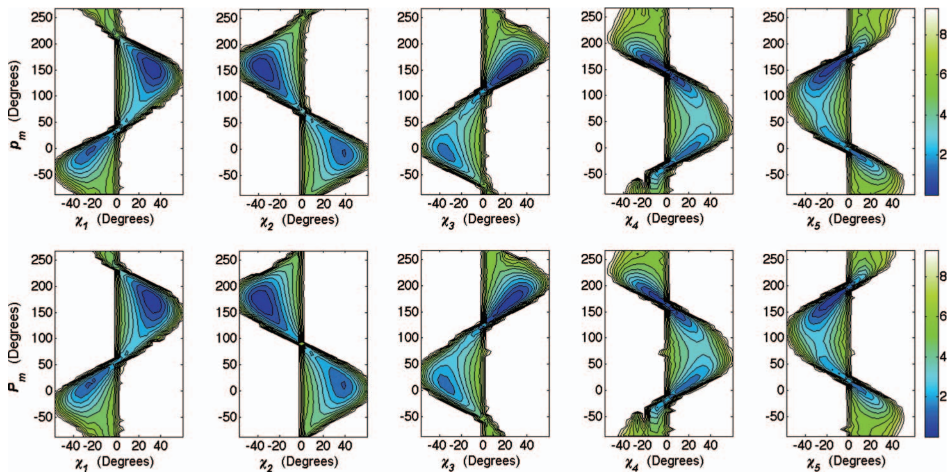

FIG. 3. Free-energy surfaces of the pseudorotation phase angle change along each χ_j pathway for *trans* proline puckering in the gas phase. The p_m values in the upper row are calculated using Eq. (3), and P_m values in the lower row are calculated using Eq. (4). The color bar has the unit of kcal/mol that applies to all the following figures as well.

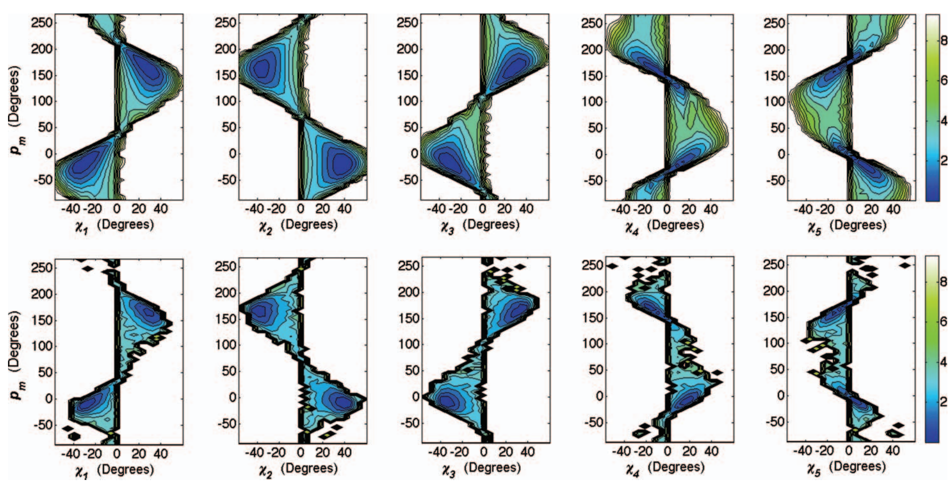

FIG. 4. Comparison of the phase angle change in each χ_j pathway of the *trans* proline puckering by the aqueous-phase (upper row) and the PDB-structure calculations (lower row). Here all p_m values are calculated using Eq. (3).

we find χ_1, χ_2 and χ_3 can clearly separate the two puckering states at $\chi_j = 0$. And among these three pathways, the phase angle (p_m or P_m) changes more symmetrically (at the negative and positive χ_j values) with χ_2 than with χ_1 or χ_3. On the other hand, χ_4 and χ_5 pathways cannot clearly separate the two puckering states at $\chi_j = 0$ since the two states both are shifted toward and to some extent across $\chi_j = 0$. Comparing the χ_j-p_m and χ_j-P_m plots, we find the p_m calculated by Eq. (3) and the P_m calculated by Eq. (4) are linearly correlated, and P_m shifts toward larger values a little bit. But the overall features described by the χ_j-p_m and χ_j-P_m plots are the same. Therefore, in the following calculations, we only use the χ_j-p_m plot.

In Fig. 4, we plot the χ_j-p_m free-energy surfaces of the *trans* proline puckering calculated by the aqueous-phase simulations and by the PDB-structure analyses. The overall trends are the same as those described by the gas-phase results except for the two differences: First, the two preferred puckering states are more equally populated; Second, p_m changes more symmetrically along the χ_2-puckering pathway. Generally, the aqueous-phase results correlate well with the PDB-structure results. In the PDB structures, the proline residues are often surrounded by proteins but not water molecules. Figure 4 shows that the proline environment in PDB structures may affect little to the five

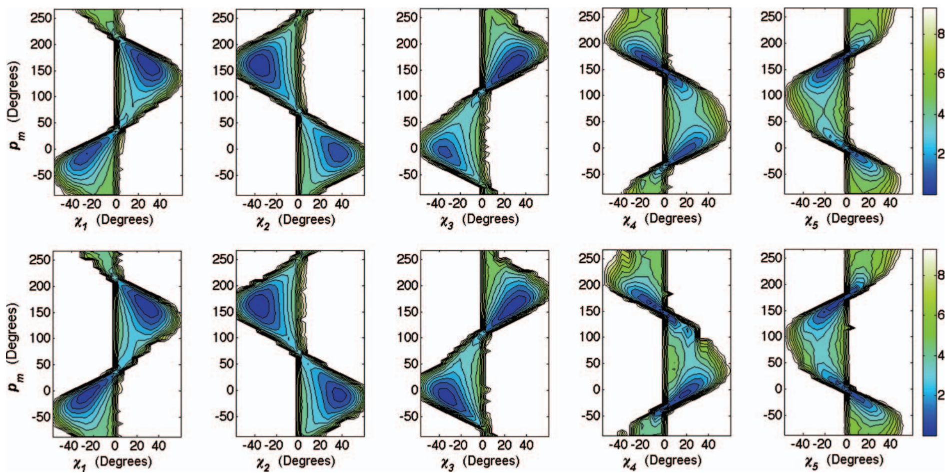

FIG. 5. The χ_j-p_m free-energy surfaces of *cis* proline puckering in the gas phase (upper row) and in water (lower row). Here all p_m values are calculated using Eq. (3).

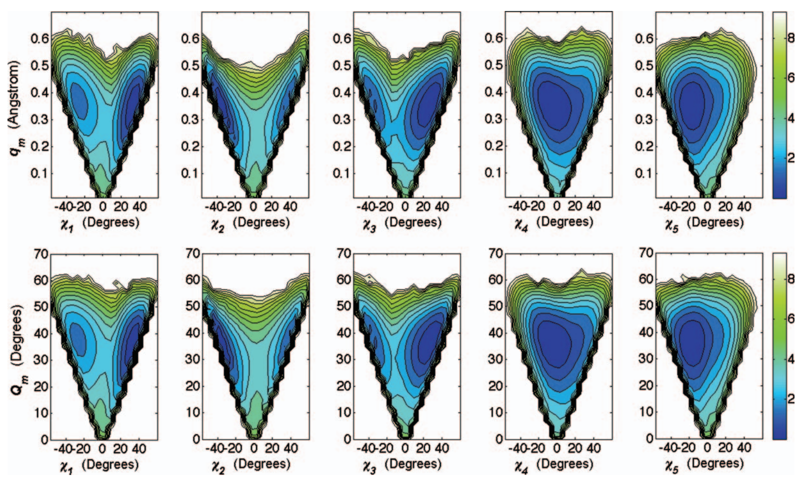

FIG. 6. Free-energy surfaces of the puckering amplitude change along each χ_j pathway of *trans* proline in the gas phase. The q_m values in the upper row are calculated using Eq. (3), and Q_m values in the lower row are calculated using Eq. (4).

χ_j puckering pathways compared with those in the aqueous environment, although their free-energy values differ slightly at the preferable puckering states.

Next, we examine the *cis* proline (ω_1 in *cis* and ω_2 in *trans* conformations) puckering pathways by the gas- and aqueous-phase simulations. We plot the corresponding χ_j-p_m free-energy surfaces in Fig. 5. Compared with Figs. 3 and 4, we find the locations of two preferred puckering states almost do not change, and the shape of the free-energy surface in each χ_j pathway resembles the corresponding one described in Figs. 3 and 4. The *trans* and *cis* proline puckering free-energy surfaces look very similar when the calculations are conducted in the aqueous phase. In Fig. 5, the results of the *cis* proline in the gas phase and *cis* proline in water also look similar except that their free-energy values at the preferred puckering states are different a little bit.

B. The relation between the torsion angle χ_j and the puckering amplitude q_m or Q_m.

Next, we examine the puckering amplitude changes along each χ_j-puckering pathway. In Fig. 6, we plot the *trans* proline free-energy surfaces of χ_j-q_m and χ_j-Q_m calculated by the gas-phase simulations. The two puckering states (described by the two dark-blue regions) separate clearly in the χ_1, χ_2 and χ_3 pathways but move closer and merge into one dark-blue region in the χ_4 and

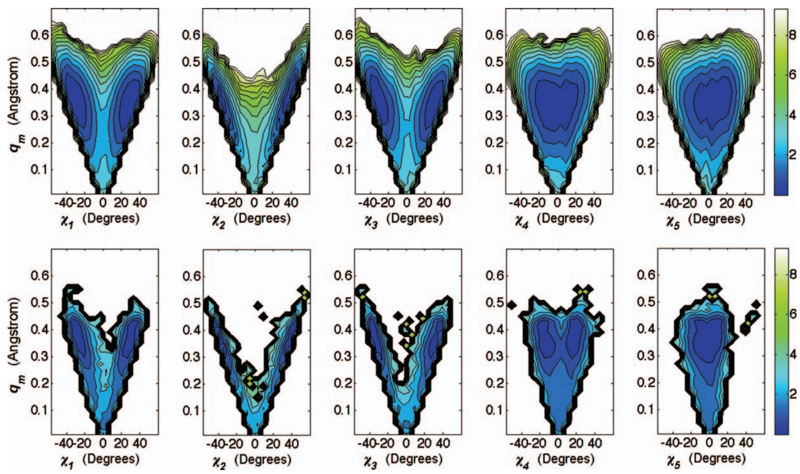

FIG. 7. Comparison of the puckering amplitude change of *trans* proline along each χ_j pathway between the aqueous-phase (upper row) and the PDB-structure calculations (lower row). All q_m values are calculated using Eq. (3).

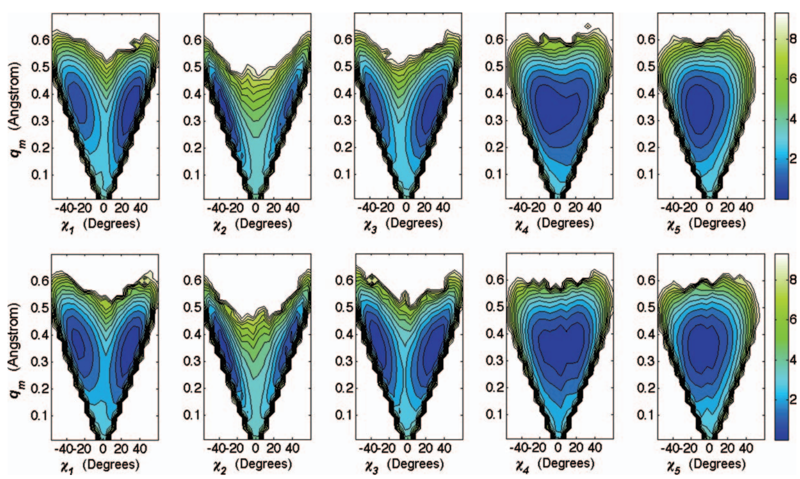

FIG. 8. The χ_j-q_m free-energy surfaces of *cis* proline puckering in the gas phase (upper row) and in water (lower row). Here all q_m values are calculated using Eq. (3).

χ_5 pathways. This shows again that χ_4 and χ_5 pathways cannot clearly distinguish proline's two preferred puckering states. In Fig. 6, we observe that the puckering amplitude (q_m or Q_m) changes almost linearly with χ_2, and its value decreases as χ_2 approaches 0° and increases otherwise. This proves that χ_2 is such a special parameter in describing proline puckering. Comparing χ_j-q_m and χ_j-Q_m subplots, we find the average values of q_m and Q_m approximately follow the relation defined by Eq. (5), and hence these subplots look very similar except for the units. Therefore, we only plot χ_j-q_m in the following calculations.

In Fig. 7, we compare the χ_j-q_m free-energy surfaces of *trans* proline puckering calculated by the aqueous-phase simulations and by the PDB-structure analyses. We observe the same trends as those described for the gas-phase results. The PDB results also verify that q_m changes more linearly along the χ_2 puckering pathway. However, the discrepancy appears in the χ_4 pathway, where the two puckering states are more clearly recognized in the PDB results than those in the aqueous-phase calculations. This may be due to the force field parameters employed.

Then we repeat the same calculations for the *cis* proline puckering pathways using the gas- and aqueous-phase simulations, respectively. We plot their χ_j-q_m free-energy surfaces in Fig. 8.

Like those observed in the χ_j-p_m plots, the shapes of the χ_j-q_m plots also look similar under the different calculation conditions (*trans* or *cis* proline conformation in either the gas phase or in water). Figure 8 shows that the puckering free-energy surfaces of the *cis* proline in the gas phase and the *cis* proline in water look similar, and they also resemble those of the *trans* proline in water (Fig. 7).

C. The five torsion angles χ_j can be categorized into two groups.

The results of Figs. 3–8 describe the direct correlation of proline puckering upon the variation of each χ_j. We find the shape of the free-energy surface along the same χ_j pathway does not change for either the *trans* or *cis* proline in either the gas phase or in water. By examining these figures, we find two typical behaviors that are represented by the χ_2 and χ_5 pathways, respectively. Therefore, we attempt to categorize the five χ_j pathways into two groups. It is obvious that χ_1, χ_2, and χ_3 can be grouped together for their similar free-energy plots. By examining Fig. 7, especially the PDB results, we find the appearance of the χ_4 pathway lies between those of the χ_2 and χ_5 pathways. However, if we look at the two puckering states at $\chi_4 \approx 0^\circ$ in Fig. 4 (PDB results), they tend to expand both the negative and positive regions of χ_4, which is more like the behavior shown in the χ_5 pathway. Therefore, we categorize the five χ_j pathways into these two groups: ***A*** (χ_1, χ_2, and χ_3) and ***B*** (χ_4 and χ_5). If we look at the geometry of the pyrrolidine ring as shown in Fig. 1, χ_4 and χ_5 lie at the two sides of the nitrogen atom. In a sense, the five torsion angles defined by the two groups distribute symmetrically around the nitrogen atom. Generally, the torsion angles in group ***A*** correlate well with the phase angle and amplitude, and the correlation is especially good in the χ_2 pathway, where χ_2 changes more symmetrically with the phase angle and linearly with the puckering amplitude. Figures 3–8 show that proline's two preferred puckering states can be conveniently symbolized using the sign of χ_j in group ***A***. In contrast, the torsion angles of group ***B*** do not show these features as clearly as those described by the χ_j of group ***A***.

D. The free-energy surface of proline puckering described by two χ_j or two τ_j.

Because proline puckering can be characterized by two parameters, we simply choose two χ_j as the reaction coordinates to describe proline puckering free-energy surfaces. In Fig. 9, we plot the results of *trans* proline puckering calculated by the gas-phase simulations in the upper triangular region; in the lower triangular region, we plot the corresponding results calculated using two τ_j that are analogous to using the triangular decomposition method.[35] All the subplots are arranged symmetrically along the diagonal line. Here each τ_j is related to each χ_j, with the definitions given in Fig. 2. Indeed, the subplot described by two χ_j resembles that described by two τ_j, and the two preferred puckering states are well characterized in each subplot.

Similarly, in Fig. 10, we plot the puckering free-energy surfaces of *trans* proline calculated by the aqueous-phase simulations in the upper triangular region, and plot those calculated by the PDB-structure analyses in the lower triangular region. Again, we observe the good correlation between the aqueous-phase and the PDB-structure calculations. Compared to the gas-phase results, the two puckering states distribute more equally as those observed in Figs. 4 and 7.

In Fig. 11, we plot the similar free-energy surfaces of the *cis* proline puckering calculated by the gas-phase simulations in the upper triangular region, and plot those calculated by the aqueous-phase simulations in the lower triangular region. The results of these two calculations look similar and they also look similar to the *trans* proline puckering results calculated by the aqueous-phase simulations (Fig. 10).

Figures 9–11 show that the free-energy surfaces described using two χ_j of group ***A*** almost follow the diagonal line, and the two puckering states can be clearly distinguished at $\chi_j = 0$. However, if using parameters of group ***B***, the two puckering states cannot be clearly distinguished at $\chi_4 = 0$ or $\chi_5 = 0$. Note that these free-energy plots can also show the relations among the five endocyclic torsion angles χ_j. For example, the subplots χ_1-χ_2 and χ_2-χ_3 show that χ_1 and χ_3 are linearly correlated with χ_2 to some extent.

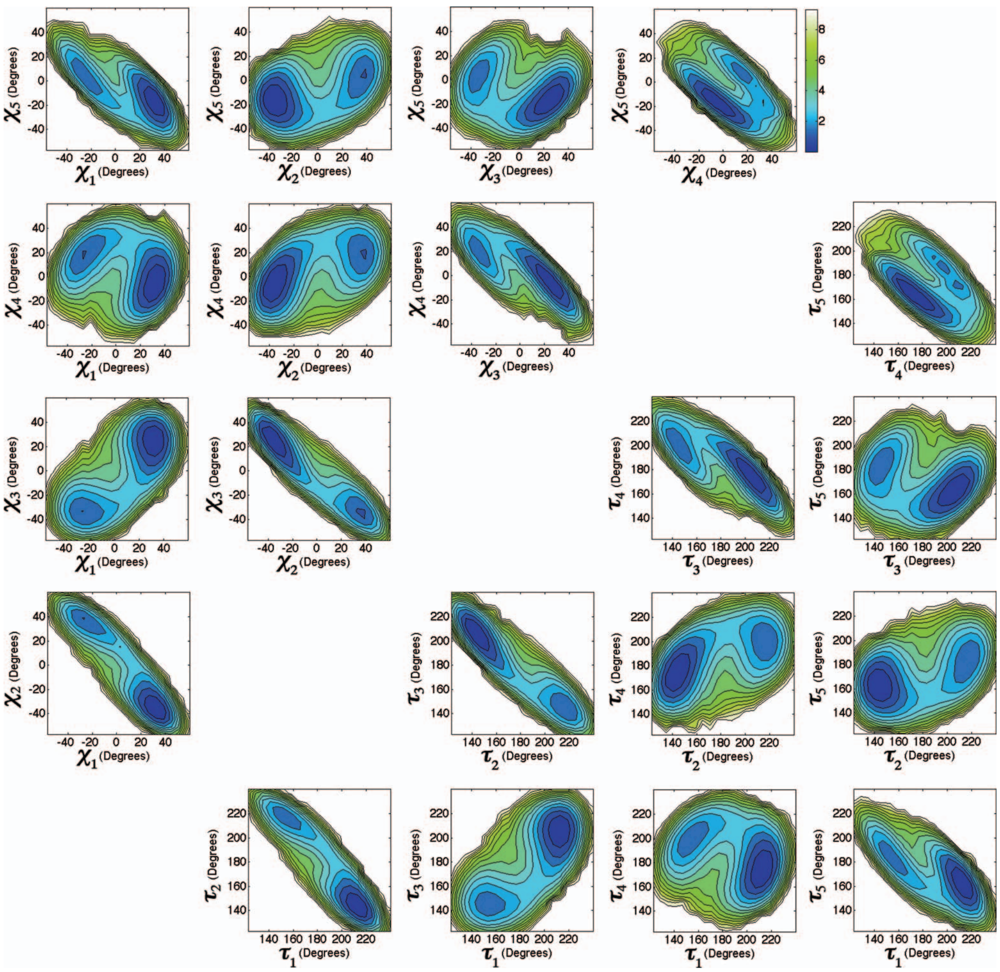

FIG. 9. The gas-phase *trans* proline puckering free-energy surfaces described by the two torsion angles χ_j and χ_k (upper triangular region), and by the two dihedral angles τ_j and τ_k (lower triangular region). Here, $\tau_j \approx \chi_j + 180^\circ$, where $\tau_j \in [0^\circ, 360^\circ]$, $\chi_j \in [-180^\circ, 180^\circ]$, and $j = 1 \sim 5$. The definitions of τ_j are given in Fig. 2.

E. Proline puckering along the backbone torsion angle ϕ.

The influence of proline puckering on the backbone torsion angle ϕ has already been noticed and studied by analyzing the PDB structures.[2–4] Here we examine ϕ-χ_j free-energy surfaces for both the *trans* and *cis* proline conformations. First, we show the free-energy surfaces of *trans* proline calculated by the gas-phase and aqueous-phase simulations and also by the PDB-structure analyses. In addition, we plot the free-energy surfaces along the ϕ coordinate using the puckering parameters p_m and q_m.

In our calculations, we choose two representative torsion angles—χ_2 of group ***A*** and χ_5 of group ***B***—as the reaction coordinates, and use them separately to examine the proline puckering upon varying the backbone torsion angle ϕ. In Fig. 12, we plot the free-energy surfaces of ϕ as a function of χ_2 or χ_5 using the gas-phase, aqueous-phase, and the PDB-structure calculations. In all the calculations, the two puckering states both are populated across a wide range of ϕ values. A common feature in Fig. 12 is that the negative-χ_2 puckering state can reach to the more negative ϕ values, and the positive-χ_2 puckering state can reach to the less negative ϕ values. The gas-phase results show that proline prefers the negative-χ_2 puckering state to the positive-χ_2 puckering state as already observed in Figs. 3, 6, and 9. The aqueous-phase results resemble those of the PDB-structure calculations. In the simulation result, we can plot a large area of free-energy surface that is not observed in the PDB result because the corresponding configurations have low probabilities, and

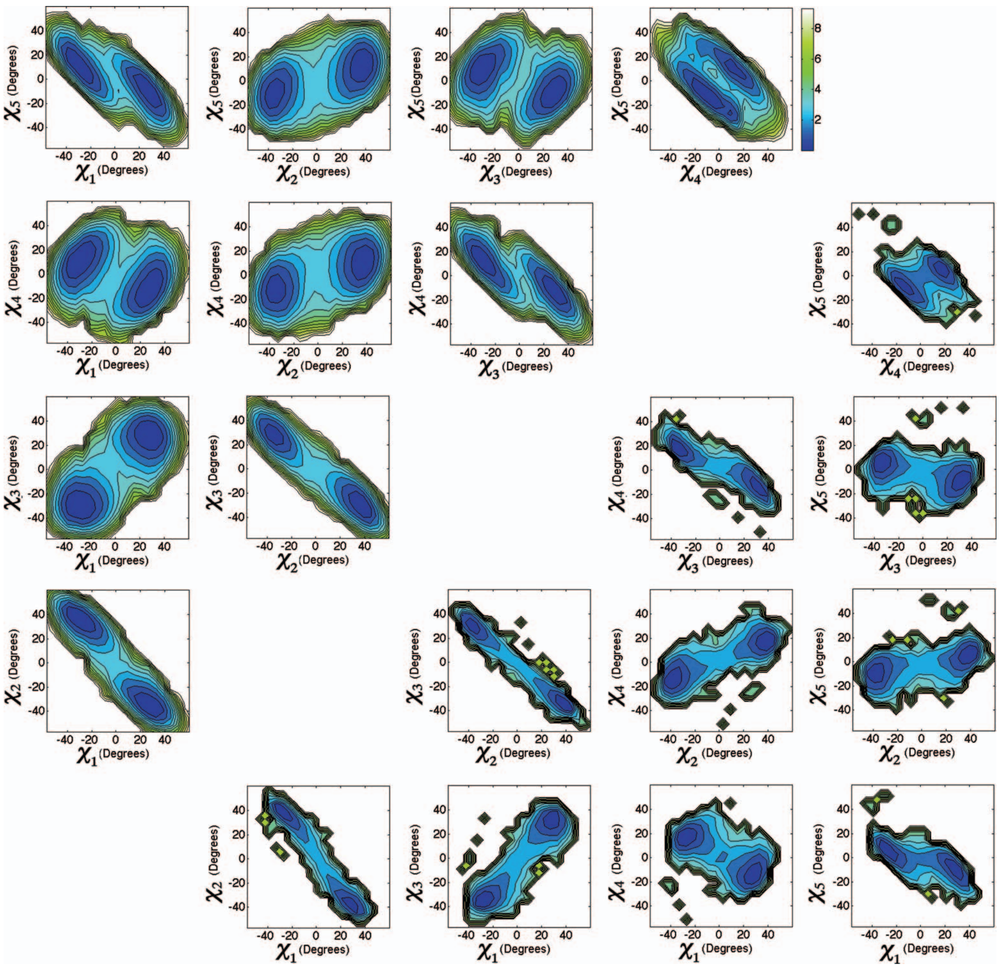

FIG. 10. The *trans* proline puckering free-energy surfaces described by the two torsion angles χ_j and χ_k from the aqueous-phase (upper triangular region) and the PDB-structure calculations (lower triangular region).

hence are less likely to be observed in the PDB structures. In the subplot of ϕ-χ_5, we observe that the correlation between ϕ and χ_5 expands a wide and stretched area, indicating the difference between ϕ and χ_5 is not small. And this was also noticed by DeTar and Luthra.[8]

To compare with the puckering pathway along the ϕ angle described by the pseudorotation phase angle and amplitude, we plot ϕ-p_m and ϕ-q_m calculated by the gas-phase and aqueous-phase simulations, and also by the PDB-structure analyses. We present the results in Fig. 13. The two preferred puckering states are well characterized in the ϕ-p_m subplot as those described by the ϕ-χ_2 subplot of Fig. 12. Because χ_2 correlates linearly with the q_m value, we can easily estimate the q_m value from the ϕ-χ_2 subplot, which can be verified using the ϕ-q_m subplot. Compared to the ϕ-p_m free-energy surface, the ϕ-χ_2 plot cannot show the phase information. However, the ϕ-χ_2 plot is sufficient to illustrate how ϕ value changes in relation to proline's two preferred puckering states.

Next, we examine ϕ-χ_j free-energy surfaces of the *cis* proline puckering in the gas phase and in water by the simulation method. We plot the results in Fig. 14. The ϕ-χ_2 plot of *cis* proline in the gas phase looks similar to that of the *cis* proline in water. Compared with Fig. 12, the results of the *trans* and *cis* proline in water look very similar.

In Fig. 15, we plot *cis* proline ϕ-p_m and ϕ-q_m free-energy surfaces calculated by the gas-phase and aqueous-phase simulations. The two-phase simulation results look similar, and they also look similar to the results of *trans* proline puckering in water (Fig. 13). The ϕ-p_m plot shows the

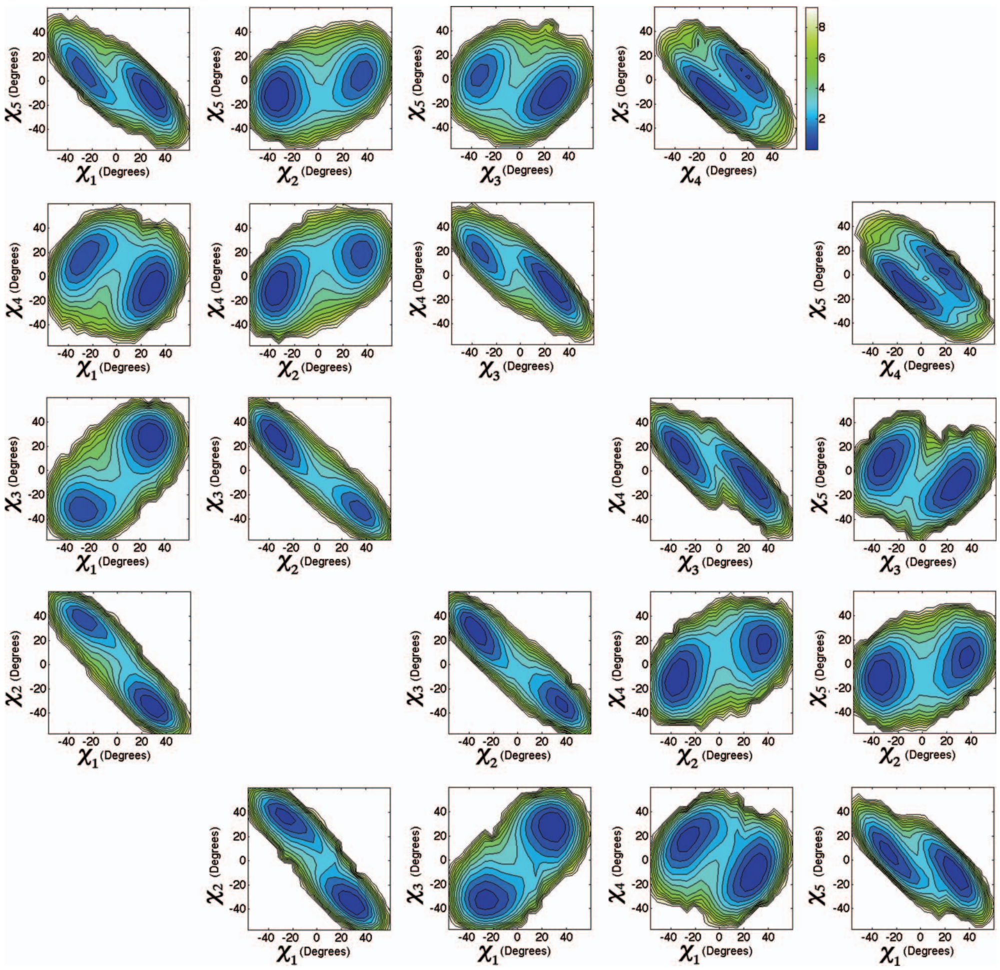

FIG. 11. The *cis* proline puckering free-energy surfaces described by the two torsion angles χ_j and χ_k from the gas-phase (upper triangular region) and the aqueous-phase calculations (lower triangular region).

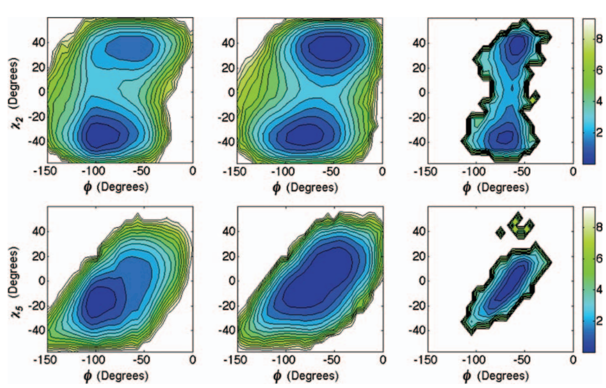

FIG. 12. The *trans* proline puckering free-energy surfaces represented by χ_2 and χ_5 upon varying the backbone torsion angle ϕ. The gas-phase, aqueous-phase, and the PDB-structure results are plotted in the left, middle, and the right columns, respectively.

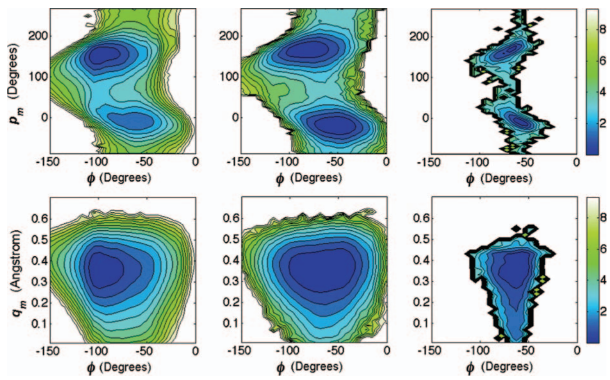

FIG. 13. The *trans* proline puckering free-energy surfaces of ϕ-p_m and ϕ-q_m using the gas-phase, aqueous-phase simulations, and the PDB-structure calculations are plotted in the left, middle, and the right columns, respectively.

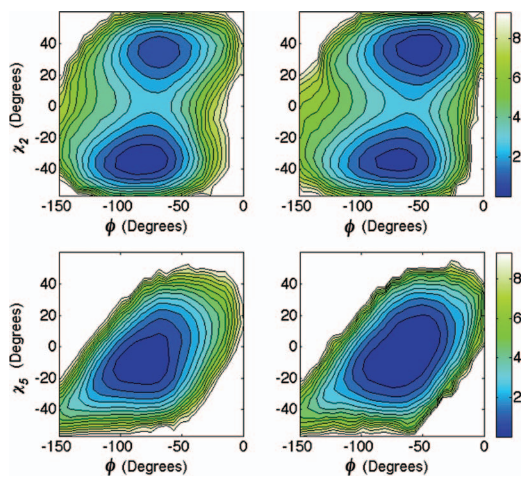

FIG. 14. The *cis* proline ϕ-χ_2 and ϕ-χ_5 free-energy surfaces by the gas-phase (left column) and the aqueous-phase (right column) calculations.

information of the two puckering states along the ϕ angle space that can also be obtained from the ϕ-χ_2 plot of Fig. 14.

F. Proline puckering along the backbone torsion angle ψ.

The correlation between proline puckering and ψ has been studied using the PDB structures along the full path of χ_1 values[2] and the partial path of χ_1 values.[3] Here we study this correlation for both the *trans* and *cis* proline conformations. First, we examine the ψ-χ_2 and ψ-χ_5 free-energy surfaces of *trans* proline puckering calculated by the gas-phase, aqueous-phase simulations and by the PDB-structure analyses. We plot the results in Fig. 16. The preferred puckering states of *trans* proline in the gas phase and in water distribute differently along the ψ angle space. In the aqueous-phase calculations, we have obtained only four preferred (ψ, χ_2) states, whereas the PDB results show an additional preferred *tCd* state at the negative-χ_2 region that has also been reported by the QM studies.[14] A common feature in Fig. 16 is that the *trans* proline prefers the negative-χ_2 pathway when ψ switches values.

Next we calculate ψ-p_m and ψ-q_m free-energy surfaces using the gas-phase, aqueous-phase simulations, and the PDB-structure analyses. We plot the results in Fig. 17. Comparing Figs. 16 and 17, the free-energy surface features of ψ-p_m plots and the ψ-χ_2 plots are analogous. This shows that

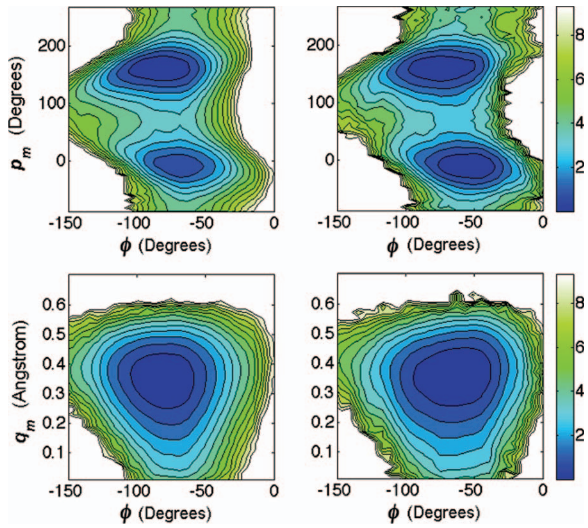

FIG. 15. The *cis* proline ϕ-p_m and ϕ-q_m free-energy surfaces by the gas-phase (left column) and the aqueous-phase (right column) calculations.

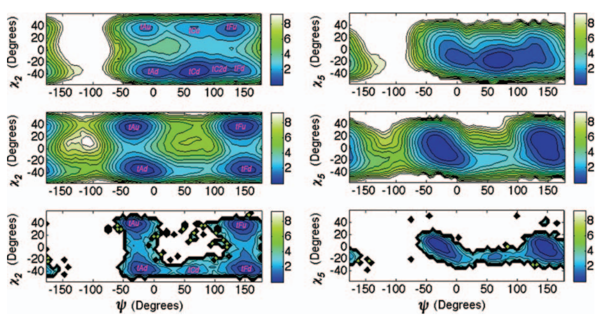

FIG. 16. The *trans* proline puckering free-energy surfaces represented by χ_2 and χ_5 upon varying the backbone torsion angle ψ. The gas-phase, aqueous-phase, and the PDB-structure results are plotted in the top, middle, and bottom rows, respectively. We have labeled seven, four, and five states in the ψ-χ_2 results by the gas-phase, aqueous-phase, and the PDB-structure calculations, respectively. These symbols are also utilized in Tables I and III. Here all the symbols except *tC2d* are taken from Ref. 14, and the letters have the following meanings: *t* denotes the *trans* conformation, *A* denotes the right-handed α helical conformation, *C* denotes the γ' conformation, *F* denotes the polyproline-like conformation, *u* denotes the UP puckering state, and *d* denotes the DOWN puckering state. The capital letters in the symbols: *A*, *C* and *F* are also utilized in Ref. 37.

the correlation between ψ and proline's two puckering states can be conveniently represented using the ψ-χ_2 plot.

Then we examine the ψ-χ_2 and ψ-χ_5 free-energy surfaces of *cis* proline puckering calculated by the gas- and aqueous-phase simulations. We plot the results in Fig. 18. Unlike the results of *trans* proline in the gas phase and in water, the free-energy surfaces of *cis* proline in the two phase calculations show similar shapes in the ψ-χ_j plots, and these shapes also resemble those of the *trans* proline in water. Although the shapes of the free-energy surfaces look similar, the free-energy values at these preferred (ψ, χ_j) states are different among the *trans* proline in water, *cis* proline in water, and *cis* proline in the gas phase.

In Fig. 19, we plot ψ-p_m and ψ-q_m free-energy surfaces of *cis* proline puckering calculated by the gas-phase and aqueous-phase simulations. Compared with Fig. 18, the ψ-p_m and ψ-χ_2 plots show similar features in the free-energy surfaces. This proves that the ψ-χ_2 plot is appropriate to use for presenting proline puckering preferences along the ψ angle space.

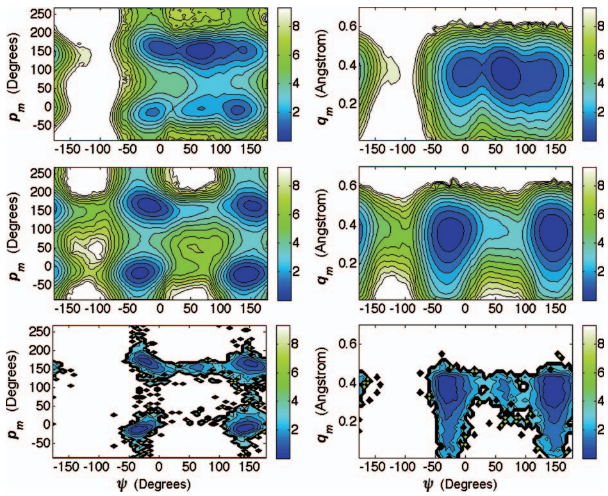

FIG. 17. The *trans* proline puckering free-energy surfaces of ψ-p_m and ψ-q_m using the gas-phase, aqueous-phase simulations, and the PDB-structure analyses are plotted in the top, middle, and bottom rows, respectively.

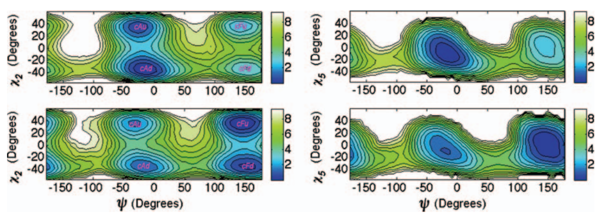

FIG. 18. The *cis* proline ψ-χ_2 and ψ-χ_5 free-energy surfaces calculated by the gas-phase (top row) and the aqueous-phase (bottom row) simulations. We have labeled four preferable states in the ψ-χ_2 free-energy surfaces. These symbols are also utilized in Tables II and IV. Here the letter c in each symbol denotes the *cis* proline conformation, and the other letters are as described in the figure caption of Fig. 16.

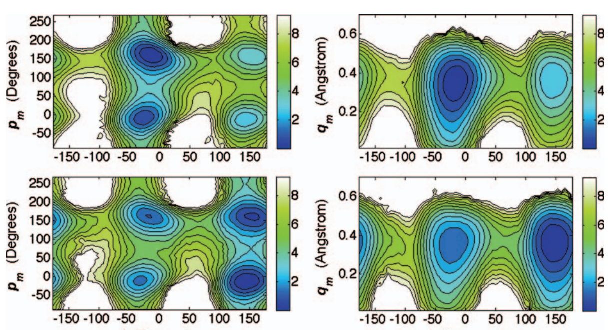

FIG. 19. The *cis* proline ψ-p_m and ψ-q_m free-energy surfaces calculated by the gas-phase (top row) and the aqueous-phase (bottom row) simulations.

G. Comparison with the QM calculations.

We have compared the results of the *trans* proline puckering free-energy surfaces by the aqueous-phase calculations with those of the PDB structure analyses. However, the proline environments are different in these two types of calculations. To validate our results, we have further compared the simulation results with those of the QM calculations and present the comparisons in Tables I to IV. In these tables, we have compared the proline conformations in the different conditions, which are the *trans* proline in the gas phase, *cis* proline in the gas phase, *trans* proline in water, and *cis* proline in water, respectively. In each table, we compare the structural parameters (ϕ, ψ, χ_j, ω_1, ω_2),

TABLE I. Structural and thermodynamic parameters compared with the QM calculations for the *trans* proline dipeptide puckering in the gas phase.

Location[a]	Method	ω_1	ϕ	ψ	ω_2	χ_1	χ_2	χ_3	χ_4	χ_5	Q_m[b]	P_m[b]	ΔF[c]	Ref.
tFu	HF[f,k]	177.0	−61.3	137.6	−175.5	−19.8	34.5	−35.7	24.8	−3.2	37.4		1.81[m]	12
	DM[d]	171	−57	123	177	−27	39	−33	21	3	41	9	1.24±0.08	TW[e]
tFd	HF[f,k]	177.6	−70.2	142.1	−176.5	30.3	−35.2	26.2	−7.3	−14.5	35.3		1.18[m]	12
	RHF[h]	176.8	−72.1	147.6	178.3	34.1	−39.1	28.8	−7.6	−16.8	39.4			13
	DM[d]	171	−75	141	177	33	−33	21	−3	−15	35	171	1.22±0.03	TW[e]
tAu	HF[f]	−171.1	−73.6	−17.7	175.4	−25.8	37.2	−34.2	18.9	4.3	38.0		3.21[m]	12, 14
	B3LYP[g]	−171.1	−77.7	−11.3	175.9	−22.7							4.01[m]	14
	RHF[h]	−175.5	−67.2	−28.8	−178.8	−29.0	38.6	−32.9	15.5	8.5	38.6			13
	RHF[i]	−171.3	−71.0	−20.9	175.4	−27.3	37.6	−33.3	16.9	6.5	38.0			13
	B3LYP[j]	−171.1	−74.5	−16.2	176.8	−25.4	36.9	−34.1	19.1	3.9	37.7			13
	DM[d]	−177	−69	−9	−177	−27	39	−33	15	9	39	9	1.93±0.07	TW[e]
tAd	HF[f,k]	−169.7	−85.1	−9.1	173.6	29.4	−38.1	31.8	−13.9	−9.7	38.1		2.11[m]	12
	RHF[h]	−171.9	−97.0	7.9	177.5	35.5	−40.3	29.5	−7.3	−18.0	40.8			13
	DM[d]	−171	−93	3	−177	33	−39	33	−9	−15	39	177	0.84±0.04	TW[e]
tCu	HF[f]	−174.7	−82.6	84.5	−174.4	−14.3	31.9	−37.1	29.7	−9.7	38.0		1.73[m]	12, 14
	B3LYP[g]	−173.9	−81.7	78.0	−175.6	−13.3	30.9	−36.5	29.7	−10.3			1.21[m]	11, 14
	RHF[h]	−175.2	−83.0	72.6	−178.8	−9.4	28.9	−37.2	33.2	−15.1	38.4			13
	RHF[i]	−175.2	−82.1	85.6	−173.7	−13.7	31.4	−37.1	30.2	−10.4	38.0			13
	B3LYP[j]	−173.0	−82.3	76.7	−175.5	−13.2	30.8	−36.7	30.0	−10.6	37.5			13
	DM[d]	−177	−75	69	177	−27	39	−39	21	3	41	15	2.06±0.07	TW[e]
tCd	HF[f]	−172.8	−86.2	74.9	−176.5	31.8	−37.5	28.2	−8.5	−14.7	37.5		0[m]	12, 14
	B3LYP[g]	−172.6	−83.6	71.2	−177.2	31.3	−37.6	28.8	−9.5	−13.7			0[m]	11, 14
	RHF[h]	−173.3	−85.3	70.0	−178.9	34.4	−39.9	29.4	−8.0	−16.8	40.0			13
	RHF[i]	−172.9	−86.1	75.8	−175.6	32.1	−37.9	28.6	−8.8	−14.7	37.9			13
	B3LYP[j]	−172.3	−84.0	70.9	−177.0	31.8	−38.2	29.3	−9.7	−14.0	38.1			13
	DM[d]	−171	−99	63	177	33	−39	27	−9	−15	35	171	0±0	TW[e]
tC2d	DM[d,l]	−177	−93	105	177	27	−33	27	−9	−15	35	171	0.67±0.03	TW[e]

[a]These symbols are also labeled in Fig. 16.
[b]To compare with the QM calculations, we only present Q_m (°) and P_m (°) results here.
[c]ΔF is the Helmholtz free energy calculated at 298 K and has the unit of kcal/mol. For the Distribution method, we present the free energy result in the form $\Delta F_{\text{avg}} \pm \Delta F_{\text{error}}$.
[d]For the Distribution method (Ref. 29), all the angles and P_m are calculated at the bin increment of 6°, and Q_m is calculated at the bin increment of 2°. So the comparison with the QM results cannot be exact.
[e]Reference from this work.
[f]HF/6-31+G(d).
[g]B3LYP/6-311++G(d,p).
[h]RHF/3−21G.
[i]RHF/6-31G(d).
[j]B3LYP/6-31G(d).
[k]The structural data are optimized at $\psi = -9.1°$, 137.6°, and 142.1° for the *tAd*, *tFu*, and *tFd* states, respectively. See footnotes of Table 1 in Ref. 12.
[l]This is only a shallow free-energy minimum between the *tCd* and *tFd* states.
[m]These are the Gibbs free energies ΔG calculated at 298.15 K.

the puckering parameters (P_m and Q_m), and the free-energy values at the different states, which are the preferred proline conformations with the lower free-energy values. Because the pyrrolidine ring puckering depends simultaneously on the backbone torsion angles, these tables provide a convenient way to show the correlations among the backbone torsion angles and χ_j.

For a quick reference, the symbols of the different states listed in the tables are also labeled in Figs. 16 and 18. Most of these symbols are taken from Ref. 14, and the capital letters in these symbols have been defined in Ref. 37. Table I lists the structural data of seven states of the *trans* proline conformations in the gas phase. The location at *tC2d* state is not obvious, even in our calculation it is only a shallow free-energy minimum between the *tCd* and *tFd* states. Note the QM methods usually

TABLE II. Structural and thermodynamic parameters compared with the QM calculations for the *cis* proline dipeptide puckering in the gas phase.

Location[a]	Method	ω_1	ϕ	ψ	ω_2	χ_1	χ_2	χ_3	χ_4	χ_5	Q_m[b]	P_m[b]	ΔF[c]	Ref.
cFu	HF[f]	−3.6	−59.1	154.1	175.7	−22.3	35.8	−35.3	22.5	−0.2	37.7		2.76[m]	12, 14
	B3LYP[g]	0.1	−62.1	146.3	177.4	−22.9							3.10[m]	14
	RHF[h]	−3.9	−65.0	176.7	178.7	−20.3	35.8	−37.3	26.2	−3.8	39.1			13
	RHF[i]	−4.6	−57.3	155.6	178.2	−23.0	36.1	−35.2	22.0	0.6	37.8			13
	B3LYP[j]	−1.2	−57.7	146.8	176.8	−25.1	36.8	−34.0	19.3	3.6	37.6			13
	DM[d]	−9	−45	141	−177	−27	39	−33	15	9	41	9	2.97±0.08	TW[e]
cFd	HF[f]	−0.5	−75.3	155.4	175.6	32.4	−36.3	25.9	−5.5	−17.0	36.7		1.9[m]	12, 14
	B3LYP[g]	0.7	−75.2	148.1	176.0	31.4							3.04[m]	14
	RHF[h]	−2.9	−70.4	174.1	179.0	34.3	−40.4	30.7	−9.4	−15.8	40.5			13
	RHF[i]	−3.4	−71.8	159.7	174.1	33.5	−37.0	26.1	−4.9	−18.1	37,6			13
	B3LYP[j]	−2.2	−74.0	143.6	175.6	32.4	−36.9	26.9	−6.6	−16.3	37.2			13
	DM[d]	−3	−69	153	−177	33	−33	27	−9	−15	35	177	3.40±0.05	TW[e]
cAu	HF[f]	5.7	−76.6	−21.9	−177.2	−23.1	36.1	−35.0	21.6	0.9	37.6		0.79[m]	12, 14
	B3LYP[g]	8.2	−79.3	−18.9	−177.3	−22.2	35.6	−35.1	22.2	0.0			1.54[m]	11, 14
	RHF[h]	9.2	−75.1	−22.3	−175.5	−26.6	38.4	−35.1	19.4	4.5	39.1			13
	RHF[i]	7.3	−76.4	−21.6	−175.7	−24.4	36.6	−34.6	20.4	2.5	37.8			13
	B3LYP[j]	11.1	−78.8	−19.0	−176.4	−24.5	36.7	−34.6	20.1	2.8	37.7			13
	DM[d]	3	−69	−27	−177	−27	39	−33	21	3	35	9	0.81±0.06	TW[e]
cAd	HF[f]	9.2	−89.5	−9.1	−179.7	31.1	−37.4	29.0	−9.8	−13.4	37.4		0[m]	12, 14
	B3LYP[g]	10.0	−91.7	−5.0	179.3	31.8	−37.5	28.4	−8.6	−14.6			0[m]	11, 14
	RHF[h]	10.8	−90.8	−1.6	−178.9	33.2	−40.1	31.4	−10.9	−14.2	40.1			13
	RHF[i]	10.7	−90.7	−5.5	−178.1	31.5	−38.1	29.8	−10.3	−13.4	38.1			13
	B3LYP[j]	11.5	−92.8	−2.2	−179.2	31.8	−38.0	29.3	−9.6	−14.1	38.0			13
	DM[d]	3	−81	−15	−177	27	−33	27	−9	−9	35	183	0±0	TW[e]

[a]These symbols are also labeled in Fig. 18.
[b]See footnote b of Table I.
[c]See footnote c of Table I.
[d]See footnote d of Table I.
[e]See footnote e of Table I.
[f]See footnote f of Table I.
[g]See footnote g of Table I.
[h]See footnote h of Table I.
[i]See footnote i of Table I.
[j]See footnote j of Table I.
[m]These are the ΔG values that are calculated by subtracting ΔG at the *cAd* state, using the results from Ref. 14

can locate three states as listed in Table I. Sahai *et al.* reported five states using the RHF/3-21G level of theory calculations.[13] Kang reported six states using the HF/6-31+G(d) level of theory calculations, where the parameters at *tAd*, *tFu*, and *tFd* were optimized at $\psi = -9.1^\circ$, 137.6°, and 142.1°, respectively.[12] For the *trans* proline in water and *cis* proline in the two phases, all the four preferable states described in the ψ-χ_2 plots are compared with the QM calculations (Tables II–IV).

In Table III (*trans* proline in water), the *tCd* state is not found by our simulation calculations, but it exists in the PDB results, and Kang has also reported this state via the QM calculations.[14] But we note that the free-energy value at the *tCd* state reported by Kang is quite large (4.46/4.13 kcal/mol), which means that this is not a favorable state compared with the other free-energy minima. Even the PDB result shows a larger free-energy value at the *tCd* state (1.41 kcal/mol). Interestingly, the *tCd* state is the most favorable state for the *trans* proline puckering in the gas phase (Table I), where the QM and the simulation results both show that the *tCd* state has the lowest free-energy value. These different features of the gas- and aqueous-phase proline puckering at the *tCd* state are manifested in Fig. 16, where the *tCd* state is outlined by the dark blue region (the lowest free-energy value) in the gas-phase results, it is missing in the aqueous-phase results, and it is represented by a light blue region (higher free-energy value) in the PDB results. The different preference for the *tCd* state of

TABLE III. Structural and thermodynamic parameters compared with the QM calculations for the *trans* proline puckering in water.

Location[a]	Method	ω_1	ϕ	ψ	ω_2	χ_1	χ_2	χ_3	χ_4	χ_5	Q_m[b]	P_m[b]	ΔF[c]	Ref.
tFu	HF[n]	−179.6	−62.6	148.8	176.4	−24.3	36.8	−34.8	20.6	2.3			0.01[r]/0[s]	14, 16
	MP2[o]	180	−58	146							41	11		17[t]
	HF[p]	−178	−63	154							38	13		17[t]
	PDB[q]	177	−57	141	177	−27	39	−33	15	3	39	9	0.08	TW[e]
	DM[d]	171	−45	141	−177	−33	39	−33	9	15	39	−3	0±0.01	TW[e]
tFd	HF[n]	−177.5	−73.5	151.6	176.6	28.8	−36.1	29.0	−11.4	−11.0			0[r]/0.23[s]	14, 16
	MP2[o]	178	−67	158							40	180		17[t]
	HF[p]	−178	−73	161							36	180		17[t]
	PDB[q]	−177	−69	153	177	27	−39	27	−15	−9	37	183	0.42	TW[e]
	DM[d]	−177	−69	153	−177	27	−33	27	−9	−9	35	183	0.61±0.13	TW[e]
tAu	HF[n]	−173.7	−68.3	−30.3	−177.6	−26.4							1.29[r]/1.43[s]	14
	PDB[q]	177	−57	−39	177	−27	39	−33	15	9	39	3	0	TW[e]
	DM[d]	−177	−63	−33	−177	−33	39	−33	15	9	39	3	0.25±0.11	TW[e]
tAd	HF[n]	−171.1	−80.8	−19.7	−179.8	27.8							1.15[r]/1.27[s]	14
	PDB[q]	−177	−63	−21	177	27	−39	33	−21	−3	35	195	0.49	TW[e]
	DM[d]	−171	−81	−21	−177	27	−33	27	−15	−9	35	183	0.11±0.07	TW[e]
tCd	HF[n]	−173.5	−86.7	85.2	−177.5	30.6							4.46[r]/4.13[s]	14
	PDB[q]	−177	−81	63	−177	33	−39	27	−9	−15	39	177	1.41	TW[e]

[a]These symbols are also labeled in Fig. 16.
[b]See footnote b of Table I.
[c]See footnote c of Table I.
[d]See footnote d of Table I.
[e]See footnote e of Table I.
[n]CPCM HF/6-31+G(d).
[o]MP2/cc-pVDZ.
[p]HF/6-31+G(d).
[q]These parameters are calculated using the configurations selected from the Protein Data Bank as described in the Method section. Similar to the DM method, all the angles and P_m are calculated using the bin increment of 6° and Q_m is calculated at the bin increment of 2°.
[r]ΔG is calculated by the CPCM HF/6-31+G(d) level of theory.
[s]ΔG is calculated by the B3LYP/6-311++G(d,p)//CPCM HF/6-31+G(d) level of theory as described in Ref. 14.
[t]In Ref. 17, Aliev and co-workers calculated these parameters for the *N*-acetyl-L-proline (AcProOH), not the proline dipeptide.

proline puckering in the gas phase and in water may be due to the intramolecular hydrogen bonding of the proline dipeptide in the gas phase that is disrupted by the interactions with the water molecules in the aqueous phase.

In the Distribution method, we have utilized the bin increment of 2° for the Q_m calculations, and the bin increment of 6° for the P_m and the angle calculations. Therefore, at least ±3° intrinsic error exists when comparing our angle data with the QM results.

V. DISCUSSION

It is interesting to note that the PDB structures encompass a more complete set of conformations along the χ_2-puckering pathway, but have few large χ-value (absolute value) conformations along the χ_1- and χ_3- pathways, and have only the medium-to-small χ-value (absolute value) conformations along the χ_4- and χ_5- pathways. These are observed in Figs. 4 and 7, where the free-energy surfaces of the aqueous-phase results are more complete than those of the PDB-structure analyses except for the χ_2-puckering pathway. This shows the special feature of χ_2 when describing proline puckering. In fact, DeTar and Luthra had already utilized χ_2 as one of the parameters in describing proline puckering.[8] In our study, we find that the χ_2-puckering pathway is linearly correlated with the puckering amplitude q_m (Figs. 6–8), and it changes more symmetrically with the phase angle p_m (Figs. 3–5). All these results demonstrate that χ_2 is a good parameter in describing proline puckering.

TABLE IV. Structural and thermodynamic parameters compared with the QM calculations for the *cis* proline puckering in water.

Location[a]	Method	ω_1	ϕ	ψ	ω_2	χ_1	χ_2	χ_3	χ_4	χ_5	Q_m[b]	P_m[b]	ΔF[c]	Ref.
cFu	HF[n]	−3.1	−61.9	153.6	175.7	−23.6	36.7	−35.3	21.5	1.3			0.14[r]/0[s]	14, 16
	MP2[o]	−2	−62	166							41	14		17[t]
	HF[p]	−2	−65	167							38	19		17[t]
	DM[d]	−9	−45	141	−177	−27	39	−33	15	9	39	9	0±0	TW[e]
cFd	HF[n]	1.6	−76.4	158.2	175.9	31.1	−36.3	27.0	−7.7	−14.8			0[r]/0.23[s]	14, 16
	MP2[o]	−6	−72	171							41	171		17[t]
	HF[p]	2	−81	171							37	172		17[t]
	DM[d]	−9	−69	153	−177	27	−33	27	−9	−9	35	183	0.41±0.04	TW[e]
cAu	HF[n]	3.4	−73.7	−22.7	−176.1	−22.0							0.97[r]/1.4[s]	14
	DM[d]	3	−63	−33	−177	−27	39	−33	15	9	39	9	1.71±0.10	TW[e]
cAd	HF[n]	9.4	−86.3	−11.3	−179.2	30.9							1.27[r]/1.95[s]	14
	DM[d]	9	−87	−15	−177	27	−33	27	−15	−9	35	183	1.27±0.11	TW[e]

[a]These symbols are also labeled in Fig. 18.
[b]See footnote b of Table I.
[c]See footnote c of Table I.
[d]See footnote d of Table I.
[e]See footnote e of Table I.
[n]See footnote n of Table III.
[o]See footnote o of Table III.
[p]See footnote p of Table III.
[t]See footnote t of Table III.
[r]ΔG is calculated by CPCM HF/6-31+G(d) level of theory, these values are calculated by subtracting the ΔG value at *cFd*, using the results from Ref. 14
[s]ΔG is calculated by B3LYP/6-311++G(d,p)//CPCM HF/6-31+G(d) level of theory, these values are calculated by subtracting the ΔG value at *cFu*, using the results from Ref. 14

When using two parameters to describe the proline puckering free-energy surfaces as those plotted in Figs. 9–11, the torsion angles in group ***A*** (χ_1, χ_2, and χ_3) can clearly separate proline's two preferred puckering states at $\chi_j = 0°$. Especially, when χ_2 is used as one of the parameters, the two puckering states are stretched along the diagonal line of the free-energy plot, making them more easily characterized. Generally, proline puckering needs to be characterized using two parameters. However, Figs. 3–11 show that we can easily distinguish proline's two preferred puckering states simply by examining the sign of χ_j in group ***A***.

Usually, proline's two puckering states both are populated across a wide region along the ϕ, ψ angle space. The ϕ-χ_j and ψ-χ_j free-energy surfaces can show not only the preferred locations of these puckering states as those described by the various QM studies (Tables I–IV), but these two-dimensional plots can also show the detailed pathways of proline puckering from one state to another. Generally, the shapes of the free-energy surfaces of *trans* proline in water, *cis* proline in water, and *cis* proline in the gas phase look very similar, and they only differ in the free-energy values at the preferred states. A common feature in Figs. 16–19 is that ψ prefers the negative-χ_2 pathway when switching values because the free-energy barrier is lower along this pathway.

VI. CONCLUSION

In this study, we have designed five proline puckering pathways using proline's five endocyclic torsion angles. We find the five puckering pathways can be categorized into two groups (χ_1, χ_2, and χ_3 in group ***A***; χ_4 and χ_5 in group ***B***). We can easily distinguish the two puckering states using the parameters of group ***A*** by examining their signs. But parameters of group ***B*** do not clearly show this feature. The torsion angle χ_2 is a good parameter in describing proline puckering because we find from our calculations that χ_2 changes linearly with the puckering amplitude q_m, and it changes more symmetrically with the phase angle p_m. This behavior of χ_2 pathway is commonly observed when we study proline dipeptide puckering under the different conditions (*trans* or *cis* conformation, in the gas phase or in water), and it is also observed in the PDB results.

The ϕ-χ_2 and ψ-χ_2 free-energy surfaces look very similar among the *trans* proline in water, *cis* proline in water, and *cis* proline in the gas phase, and they differ only in the free-energy values at the preferred states. The ψ-χ_2 free-energy surfaces also show that ψ prefers the negative-χ_2 pathway when switching values.

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